Configurations of heavy tetraquark

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莊澤堃

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Motivation

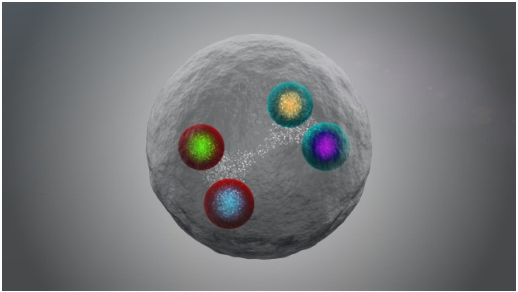

Figure 1: Charmed existence: conceptual illustration of a tetraquark. (Courtesy: CERN)

Hamiltonian for tetraquark

$$H = \sum_{1}^{4} m_i + \sum_{i<j} V_{ij}(r_{ij}) \tag{1}$$

$$V_{ij}(r_{ij}) = \lambda_i \cdot \lambda_j A_{ij} + B_{ij} \frac{1}{m_i m_j} s_i \cdot s_j \lambda_i \cdot \lambda_j$$

Diquark-antiquark configuration

Color-wave function

With this configuration, the flavor-wasve function is a trival symmetric function. The color space acting on $\lambda_i^a \lambda_j^a$ in a given flavor configuration of tetraquark can be decomposed using the irreducible representation of color SU(3) as

$$3 \otimes 3 \otimes \bar{3} \otimes \bar{3} = \bar{3} \otimes 3 + 6 \otimes \bar{6} + \bar{3} \otimes 3 + 6 \otimes 3. \tag{2}$$

Color singlet states are given by the first and scond term in the right-hand side of Eq.(2). The correseponding wave function can be constructed following:

$$|\zeta_1\rangle = (Q_1 Q_2)^6 \otimes (\bar{Q}_3 \bar{Q}_4)^{\bar{6}} \qquad |\zeta_2\rangle = (Q_1 Q_2)^{\bar{3}} \otimes (\bar{Q}_3 \bar{Q}_4)^3$$

Spin-wave function

The spin space of tetraquark with diquark-antidiquark configuration can be expressed as $V_{\frac{1}{2}} \otimes V_{\frac{1}{2}} \otimes V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}$, decomposed into the direct sum of the following parts:

$$V_{1/2} \otimes V_{1/2} \otimes V_{1/2} \otimes V_{1/2} = V_0 \otimes V_0 \oplus V_0 \otimes V_1 \oplus V_1 \otimes V_0 \oplus V_1 \otimes V_1 \tag{3}$$

- Total spin $S = 0$ from $V_0 \otimes V_0$

The correseponding bases are given following:

$$\chi_0^{00} = \frac{1}{\sqrt{2}} |\uparrow\downarrow - \downarrow\uparrow\rangle \otimes \frac{1}{\sqrt{2}} |\uparrow\downarrow - \downarrow\uparrow\rangle \tag{4}$$

$$\chi_0^{11} = \frac{1}{\sqrt{3}} [|\uparrow\uparrow\rangle \otimes |\downarrow\downarrow\rangle] + \frac{1}{\sqrt{3}} [|\downarrow\downarrow\rangle \otimes |\uparrow\uparrow\rangle] - \frac{1}{\sqrt{3}} \left[\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \otimes \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)\right] \tag{5}$$

Spin-wave function

Here, we define the spinors:

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{6}$$

- $S = 1$ from $V_1 \otimes V_1$, $V_1 \otimes V_0$, $V_0 \otimes V_1$

$$|\chi_1^{11}\rangle = \frac{1}{\sqrt{2}}\left[\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \otimes |\uparrow\uparrow\rangle\right] - \frac{1}{\sqrt{2}}\left[|\uparrow\uparrow\rangle + \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)\right] \tag{7}$$

$$|\chi_1^{10}\rangle = |\uparrow\uparrow\rangle \otimes \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \text{ (Antisymmetry)} \tag{8}$$

$$|\chi_1^{01}\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \otimes |\uparrow\uparrow\rangle \text{ (Antisymmetry)} \tag{9}$$

Spin-wave function

- $S = 2$ from $V_1 \otimes V_1$

$$\left|\chi_2^{11}\right\rangle = |\uparrow\uparrow\rangle \otimes |\uparrow\uparrow\rangle \tag{10}$$

W. Park and S. H. Lee, Nucl. Phys. A **925**, 161-184 (2014) [arXiv:1311.5330 [nucl-th]].

M. S. Liu, Q. F. Lü, X. H. Zhong and Q. Zhao, Phys. Rev. D **100**, no.1, 016006 (2019) [arXiv:1901.02564 [hep-ph]].

The matrix elements of $S_i \cdot S_j$

$S_i \cdot S_j$	$\left\{\left\|\chi_0^{11}\right\rangle, \left\|\chi_0^{00}\right\rangle\right\}$	$\left\{\left\|\chi_2^{11}\right\rangle\right\}$	$\left\{\left\|\chi_1^{10}\right\rangle, \left\|\chi_1^{11}\right\rangle, \left\|\chi_1^{01}\right\rangle\right\}$
$S_1 \cdot S_2$	$\begin{pmatrix} 1/4 & 0 \\ 0 & -3/4 \end{pmatrix}$	$1/4$	$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & -3/4 \end{pmatrix}$
$S_1 \cdot S_3$	$\begin{pmatrix} -1/2 & -\sqrt{3}/4 \\ -\sqrt{3}/4 & 0 \end{pmatrix}$	$1/4$	$\begin{pmatrix} 0 & -\sqrt{2}/4 & 1/4 \\ -\sqrt{2}/4 & 1/4 & \sqrt{2}/4 \\ -1/4 & \sqrt{2}/4 & 0 \end{pmatrix}$
$S_1 \cdot S_4$	$\begin{pmatrix} -1/2 & \sqrt{3}/4 \\ \sqrt{3}/4 & 0 \end{pmatrix}$	$1/4$	$\begin{pmatrix} 0 & \sqrt{2}/4 & -1/4 \\ \sqrt{2}/4 & -1/4 & \sqrt{2}/4 \\ -1/4 & \sqrt{2}/4 & 0 \end{pmatrix}$
$S_2 \cdot S_3$	$\begin{pmatrix} -1/2 & \sqrt{3}/4 \\ \sqrt{3}/4 & 0 \end{pmatrix}$	$1/4$	$\begin{pmatrix} 0 & -\sqrt{2}/4 & -1/4 \\ -\sqrt{2}/4 & -1/4 & -\sqrt{2}/4 \\ -1/4 & -\sqrt{2}/4 & 0 \end{pmatrix}$
$S_2 \cdot S_4$	$\begin{pmatrix} -1/2 & -\sqrt{3}/4 \\ -\sqrt{3}/4 & 0 \end{pmatrix}$	$1/4$	$\begin{pmatrix} -3/4 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1/4 \end{pmatrix}$
$S_3 \cdot S_4$	$\begin{pmatrix} 1/4 & 0 \\ 0 & -3/4 \end{pmatrix}$	$1/4$	$\begin{pmatrix} -3/4 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1/4 \end{pmatrix}$

Table 1: Matrix elements of $S_i \cdot S_j$

The matrix elements of $\lambda_i^a \lambda_j^a$

$\lambda_i^a \lambda_j^a$	$(q_1 q_2)^6 (\bar{q}\bar{q})^{\bar{6}}$, $(q_1 q_2)^{\bar{3}} (\bar{q}\bar{q})^3$	$(q_1 \bar{q}_3)^8 (q_2 \bar{q}_4)^8$, $(q_1 \bar{q}_3)^1 (\bar{q}\bar{q})^1$
$\lambda_1^a \lambda_2^a = \lambda_3^a \lambda_4^a$	$\begin{pmatrix} 4/3 & 0 \\ 0 & -8/3 \end{pmatrix}$	$\begin{pmatrix} -4/3 & 4\sqrt{2}/3 \\ 4\sqrt{2}/3 & 0 \end{pmatrix}$
$\lambda_1^a \lambda_3^a = \lambda_2^a \lambda_4^a$	$\begin{pmatrix} -10/3 & -2\sqrt{2} \\ -2\sqrt{2} & -4/3 \end{pmatrix}$	$\begin{pmatrix} 2/3 & 0 \\ 0 & -16/3 \end{pmatrix}$
$\lambda_1^a \lambda_4^a = \lambda_2^a \lambda_3^a$	$\begin{pmatrix} -10/3 & 2\sqrt{2} \\ 2\sqrt{2} & -4/3 \end{pmatrix}$	$\begin{pmatrix} -14/3 & -4\sqrt{2}/3 \\ -4\sqrt{2}/3 & 0 \end{pmatrix}$

Table 2: The matrix elements of $\lambda_i^a \lambda_j^a$

Wave function of tetraquark with diquark-antidiquark configuration

The total wave function of the tetraquark is the direct product of space, flavor, color and spin wave function, i.e.,

$$|\psi\rangle = \left|\psi_{\text{space}}\right\rangle |\psi_{\text{flavor}}\rangle |\psi_{\text{color}}\rangle \left|\psi_{\text{spin}}\right\rangle . \tag{11}$$

Since we consider the groud state of tetraquark, the spatial wave function is symmetry. The configurations classified by J^{PC} within Pauli exclusion principle are listed following.

Wave function of tetraquark with diquark-antidiquark configuration

	Existence	Non-eistence
0^{++}	$\left\lvert (cc)^{6}_{0}(\bar{c}\bar{c})^{\bar{6}}_{0}\right\rangle_{0}, \left\lvert (cc)^{\bar{3}}_{1}(\bar{c}\bar{c})^{3}_{1}\right\rangle_{0}$	—
1^{+-}	$\left\lvert (cc)^{\bar{3}}_{1}(\bar{3}\bar{3})^{3}_{1}\right\rangle_{1}$	
2^{++}	$\left\lvert (cc)^{\bar{3}}_{1}(\bar{c}\bar{c})^{3}_{1}\right\rangle_{2}$	—

Table 3: Configurations of tetraquark with different J^{PC}

The mass splitting in hyperfine potential

- 0^{++}

$$\begin{pmatrix} \langle \zeta_1 | \lambda_i \cdot \lambda_j | \zeta_1 \rangle \langle \chi_0^{00} | \chi_0^{00} \rangle & \langle \zeta_1 | \lambda_i \cdot \lambda_j | \zeta_2 \rangle \langle \chi_0^{00} | \chi_0^{11} \rangle \\ \langle \zeta_2 | \lambda_i \cdot \lambda_j | \zeta_1 \rangle \langle \chi_0^{11} | \chi_0^{00} \rangle & \langle \zeta_2 | \lambda_i \cdot \lambda_j | \zeta_2 \rangle \langle \chi_0^{11} | \chi_0^{11} \rangle \end{pmatrix} \mathcal{A}_{ij} \tag{12}$$

$$\frac{1}{m_i m_j} \begin{pmatrix} \langle \zeta_1 | \lambda_i \cdot \lambda_j | \zeta 1 \rangle \langle \chi_0^{00} | S_i \cdot S_j | \chi_0^{00} \rangle & \langle \zeta_1 | \lambda_i \cdot \lambda_j | \zeta_2 \rangle \langle \chi_0^{00} | S_i \cdot S_j | \chi_0^{11} \rangle \\ \langle \zeta_2 | \lambda_i \cdot \lambda_j | \zeta_1 \rangle \langle \chi_0^{11} | S_i \cdot S_j | \chi_0^{00} \rangle & \langle \zeta_2 | \lambda_i \cdot \lambda_j | \zeta_2 \rangle \langle \chi_0^{11} | S_i \cdot S_j | \chi_0^{11} \rangle \end{pmatrix} \mathcal{B}_{ij} \tag{13}$$

The mass splitting in hyperfine potential

$$V^{0^{++}} = (\mathcal{A}_{12} + \mathcal{A}_{34}) \begin{pmatrix} 4/3 & 0 \\ 0 & -8/3 \end{pmatrix} + (\mathcal{A}_{13} + \mathcal{A}_{24} + \mathcal{A}_{14} + \mathcal{A}_{23}) \begin{pmatrix} -10/3 & 0 \\ 0 & -4/3 \end{pmatrix}$$
$$+ \frac{\mathcal{B}_{12} + \mathcal{B}_{34}}{m_c m_c} \begin{pmatrix} -1 & 0 \\ 0 & -2/3 \end{pmatrix} + \frac{\mathcal{B}_{13} + \mathcal{B}_{24} + \mathcal{B}_{14} + \mathcal{B}_{23}}{m_{\bar{c}} m_{\bar{c}}} \begin{pmatrix} 0 & \sqrt{6}/2 \\ \sqrt{6}/2 & 2/3 \end{pmatrix} \quad (14)$$

- 1^{+-}

$$V^{1^{+-}} = (\mathcal{A}_{12} + \mathcal{A}_{34}) \left(-\frac{8}{3}\right) + (\mathcal{A}_{13} + \mathcal{A}_{24} + \mathcal{A}_{14} + \mathcal{A}_{23}) \left(-\frac{4}{3}\right)$$
$$+ \frac{\mathcal{B}_{12} + \mathcal{B}_{34}}{m_c m_c} \left(-\frac{2}{3}\right) + \frac{(\mathcal{B}_{13} + \mathcal{B}_{24} + \mathcal{B}_{14} + \mathcal{B}_{23})}{m_{\bar{c}} m_{\bar{c}}} \left(\frac{2}{3}\right) \quad (15)$$

The mass splitting in hyperfine potential

- 2^{++}

$$V^{2++} = (\mathcal{A}_{12} + \mathcal{A}_{34})\left(-\frac{8}{3}\right) + (\mathcal{A}_{13} + \mathcal{A}_{24} + \mathcal{A}_{14} + \mathcal{A}_{23})\left(-\frac{4}{3}\right) + \frac{\mathcal{B}_{12} + \mathcal{B}_{34}}{m_c m_c}\left(-\frac{2}{3}\right) + \frac{\mathcal{B}_{13} + \mathcal{B}_{24} + \mathcal{B}_{14} + \mathcal{B}_{23}}{m_{\bar{c}} m_{\bar{c}}}\left(-\frac{1}{3}\right) \quad (16)$$

Extract $\mathcal{A}_{13}$ and $\mathcal{B}_{13}$

The Hamiltonian for charmonium is given by

$$H_{c\bar{c}} = \left(\sum_{i=1}^{2} m_i + T_i \right) + V_{13}, \tag{17}$$

where $V_{13} = \lambda_1 \cdot \lambda_3 A(r_{13}) + \frac{B(r_{13})}{m_1 m_3} S_1 \cdot S_3 \lambda_1 \cdot \lambda_3$. Within bases $\left\{ |c\bar{c}\rangle_0^1,\ |c\bar{c}\rangle_1^1 \right\}$, we have

$$M_{\eta_c} = -\frac{16}{3}\mathcal{A}_{13} + 4\mathcal{B}_{13}\frac{1}{m_c^2} + 2m_c \tag{18}$$

$$M_{J/\psi} = -\frac{16}{3}\mathcal{A}_{13} - \frac{4}{3}\mathcal{B}_{13}\frac{1}{m_c^2} + 2m_c \tag{19}$$

Extract $\mathcal{A}_{12}$ and $\mathcal{B}_{12}$

$\mathcal{A}_{12}$ and $\mathcal{B}_{12}$ can be extracted from Hamiltonian governing doubly heavy baryon with components $c(1)c(2)q(3)$. The spin of component $c(1)c(2)$ only is $S_{cc} = 1$ because the irreducible color representation $\bar{3}$ is antisymmetry. So the spin of double baryons we consider are $S = \frac{3}{2}$ and $S = \frac{1}{2}$. The spin eigenstates of ccq can be written as

$$\left|(cc)_1, q_{\frac{1}{2}}; \frac{3}{2}\right\rangle, \left|(cc)_1, q_{\frac{1}{2}}; \frac{1}{2}\right\rangle. \tag{20}$$

The corresponding Hamiltonian is

$$H = \sum_{i=1}^{3} (m_i + T_i) + A_{12}\lambda_1 \cdot \lambda_2 + A_{13}\lambda_1 \cdot \lambda_3 + A_{23}\lambda_2 \cdot \lambda_3 + \frac{B_{12}}{m_c m_c}\lambda_1 \cdot \lambda_2 S_1 \cdot S_2 + \frac{B_{13}}{m_c m_q}\lambda_1 \cdot \lambda_3 S_1 \cdot S_3 + \frac{B_{23}}{m_c m_q}\lambda_2 \cdot \lambda_3 S_2 \cdot S_3 \tag{21}$$

Extract $\mathcal{A}_{12}$ and $\mathcal{B}_{12}$

For baryon, the color singlet function is

$$\zeta = \frac{1}{\sqrt{6}}(rgb - rbg + gbr - grb + brg - bgr) \tag{22}$$

L. Y. Xiao and X. H. Zhong, [arXiv:1302.0079 [hep-ph]].

The matrix elements $\langle \lambda_i \cdot \lambda_j \rangle$ in this basis are given following:

$$\langle \lambda_1 \cdot \lambda_2 \rangle = \langle \lambda_1 \cdot \lambda_3 \rangle = \langle \lambda_2 \cdot \lambda_3 \rangle = \frac{-8}{3}.$$

However, the bases Eq.(20) are not suitable for calculating the expectations of $S_1 \cdot S_3$ and $S_2 \cdot S_3$, because The paremeters $A_{13}(B_{13})$ and $A_{23}(B_{23})$ stand for the couple of $c(1)q$ and $c(2)q$, respectively.

Extract $\mathcal{A}_{12}$ and $\mathcal{B}_{12}$

A change from basis $\left|(S_c, S_c)S_{cc}, S_q; S\right\rangle$ to $\left|S_c, (S_c, S_q)S\right\rangle$ can be accomplished in following way:

$$\left|S_c, (S_c, S_q)S_{cq}; S\right\rangle = \sum_{S_{cc}=|S_c-S_c|}^{S_c+S_c} (-1)^{S_c+S_c+S_q+S}\sqrt{2S_{cc}+1}\sqrt{2S_{cq}+1}\times \begin{Bmatrix} S_c & S_c & S_{cc} \\ S_q & S & S_{cq} \end{Bmatrix} \left|(S_c, S_c)S_{cc}, S_q; S\right\rangle \quad (23)$$

For $\left|(S_{c2}, S_{c1})S_{cc}, S_q; S\right\rangle$, we have the relationship:

$$\begin{aligned} \left|(S_{c1}, S_{c2})S_{cc}, S_q; S\right\rangle &= (-1)^{S_{c1c2}-S_{c1}-S_{c2}} \left|(S_{c2}, S_{c1})S_{cc}, S_q; S\right\rangle \\ &= \left|(S_{c2}, S_{c1})S_{cc}, S_q; S\right\rangle \end{aligned}$$

Extract $\mathcal{A}_{12}$ and $\mathcal{B}_{12}$

With Eq.(23),

$$\left|(cc)_1, q_{\frac{1}{2}}; \frac{3}{2}\right\rangle = \left|c_{\frac{1}{2}}, (cq)_1; \frac{3}{2}\right\rangle \tag{24}$$

$$\left|(cc)_1, q_{\frac{1}{2}}; \frac{1}{2}\right\rangle = \frac{\sqrt{3}}{2}\left|c_{\frac{1}{2}}, (cq)_0; \frac{1}{2}\right\rangle + \frac{1}{2}\left|c_{\frac{1}{2}}, (cq)_1; \frac{1}{2}\right\rangle \tag{25}$$

Extract $\mathcal{A}_{12}$ and $\mathcal{B}_{12}$

$S_i \cdot S_j$	$\left\lvert (cc)_1, q_{\frac{1}{2}}; \frac{3}{2} \right\rangle$	$\left\lvert c_{\frac{1}{2}}, (cq)_1; \frac{3}{2} \right\rangle$	$\left\lvert c_{\frac{1}{2}}, (cq)_1; \frac{1}{2} \right\rangle$	$\left\lvert c_{\frac{1}{2}}, (cq)_0; \frac{1}{2} \right\rangle$
$S_1 \cdot S_2$	$\frac{1}{4}$	—	—	—
$S_1 \cdot S_3$	—	$\frac{1}{4}$	$\frac{1}{4}$	$-\frac{3}{4}$
$S_2 \cdot S_3$	—	$\frac{1}{4}$	$\frac{1}{4}$	$-\frac{3}{4}$

Table 4: The expectations of $S_i \cdot S_j$

Extract $\mathcal{A}_{12}$ and $\mathcal{B}_{12}$

We adopt the masses of doubly heavy baryons Ω_{cc} and Ω^*_{cc}.

Parameters	Values
A12	120.31944
B12	-14250.0
A13	-0.01287
B13	-0.04767

Table 5: Parameters

N. Mathur and M. Padmanath, Phys. Rev. D **99**, no.3, 031501 (2019) doi:10.1103/PhysRevD.99.031501 [arXiv:1807.00174 [hep-lat]].

The codes for these calculations can be downloaded from this URL: `https://github.com/PLHzzj/zhang.git`

Mass spectra

0^{++}	0^{++}	1^{+-}	2^{++}
6098.14	6192.89	6192.89	6277.64

Table 6: Mass spectra (unit: MeV)